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BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

FIRST SEMESTER, 2023-2024

CE-F213 SURVEYING

Lab Quiz

Time: 10 minutes

Date: 22/11/2023

Maximum Marks: 20

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1. When a total station is cited to the target, which operation acts first?

☒ (a) Rotation of the optical axis

(c) Rotation of the horizontal axis

(b) Rotation of the Vertical axis

(d) Rotation of line of collimation

2. The length of a chain is measured from:

☒ (a) Centre of one handle to the centre of the other handle

☒ (b) Outside of one handle to outside of the other handle

(c) Outside of one handle to the inside of the other handle

(d) Inside of one handle to the inside of the other handle

3. The smallest graduated division in leveling staff is 5 mm

4. The size of a theodolite is determined using:

☒ (a) Diameter of upper plate

☒ (b) Diameter of lower plate

(c) Both the upper and lower plate

(d) All of the above

5. The least count of theodolite is 20 minute for Main Scale
20 seconds for Vernier Scale

6. Height of Instrument (HI) =

☒ (a) Reduced level (R.L) + Back sight (B.S)

(b) Reduced level (R.L) + Foresight (F.S)

(c) Reduced level (R.L) + Benchmark (B.M)

(d) Reduced level (R.L) + Intermediate sight (I.S)

7. What is the value of the back bearing of line AB when the fore bearing of the line is 165° ?

(a) 180°

(b) 75°

☒ (c) 345°

(d) 65°

8. Calculate the length of the chord if the radius is 30m and the angle of deflection is 60°

(a) 20m

(b) 25m

☒ (c) 30m

(d) 35m

9. If the magnetic declination is 4°E and the magnetic bearing of a line is 59° , the true bearing is:

- (a) 63° (c) 67°
(b) 55° (d) 51°

10. The instrument station is setup at a station and rays are drawn to various stations to be plotted. This method of plane table surveying is known as:

- (a) Intersection (c) Traversing
(b) Radiation (d) Resection

11. The chainage at point of intersection is 756.45 m, radius of curve is 75 m and the intersection angle is 60° . The chainage at point of curve is:

- (a) 626.55m (c) 598.78m
(b) 713.58m (d) 636.67m

12. Setting out of simple curve by offset from chord produced method is suitable when:

- (a) Theodolite is not available (c) When radius of curve is minimum
(b) Length of curve is minimum (d) All of the above

13. The beveled edge of the alidade in plane table surveying is also known as

Fiducial Edge

14. Convert the following whole circle bearings to the quadrantal bearings:

140° 40° East of South E 40° S
 339° 21° West of North W 21° N

15. Axis of rotation of theodolite in vertical plane indicates Horizontal Plane.

16. The obstacle, which obstructs vision but not chaining, is

- (a) River (c) Hill
(b) Pond (d) All of these

17. In the prismatic compass

- (a) the magnetic needle moves with the box
(b) the line of the sight does not move with the box
(c) the magnetic needle and graduated circle do not move with the box
(d) the graduated circle is fixed to the box and the magnetic needle always remains in the N-S direction

18. The R.L. of the point A which is on the floor is 100 m and back sight reading on A is 2.455 m. If the foresight reading on the point B which is on the ceiling is 2.745 m, the R.L. of point B will be:

(a) 94.80 m

(b) 99.71 m

(c) 100.29 m

☒ (d) 105.20 m

19. The instrument used for accurate centering in plane table survey is:

(a) Spirit level

(b) Alidade

☒ (c) Plumbing fork

(d) Trough compass

20. If the lower clamp screw is tightened and upper clamp screw is loosened, the theodolite may be rotated:

(a) on its outer spindle with a relative motion between the vernier and graduated scale of lower plate

(b) on its outer spindle without a relative motion between the vernier and graduated scale of lower plate

☒ (c) on its inner spindle with a relative motion between the vernier and the graduated scale of lower plate

(d) on its inner spindle without a relative motion between the vernier and the graduated scale of lower plate